Plant Equipment — Troubleshooting Guide

Covers: Pumps, Compressors, Rotating Equipment

1. Using This Guide

This guide lists symptoms observed on rotating equipment (pumps and compressors) together with likely root causes and the recommended first diagnostic action. It is intended to be used together with the equipment-specific manual and current sensor readings from the CMMS.

2. Symptom: Abnormal Vibration

Vibration Pattern	Likely Cause	First Action
1x running speed, steady	Rotor/impeller imbalance	Inspect impeller, check for debris/erosion
2x running speed	Misalignment	Perform laser alignment check
High frequency, broadband	Bearing defect	Inspect bearing, check lubrication
Erratic / crackling noise	Cavitation	Check suction pressure and NPSH
Sudden spike after startup	Loose foundation bolts	Inspect baseplate and anchor bolts

3. Symptom: Overheating

Bearing overheating without a strong vibration change usually points to a lubrication problem: check oil level and condition first, then re-grease or replace lubricant per the equipment manual. Motor overheating combined with reduced flow output suggests the equipment may be running below its minimum continuous flow, which should be verified against the process setpoint. Persistent overheating after lubrication service should trigger an alignment check, since misalignment increases radial bearing load and generates heat even when vibration readings remain moderate.

4. Symptom: Pressure Deviation

Low discharge pressure with normal motor current typically indicates a clogged suction strainer, a partially closed suction valve, or internal wear (worn wear rings, impeller erosion) reducing pump efficiency. Low pressure combined with high motor current can indicate a partially blocked discharge line. High discharge pressure with reduced flow can indicate a downstream blockage or a closed/failed discharge valve.

5. When to Escalate

Any reading beyond the alarm thresholds defined in the equipment manual should result in a work order being raised. If a safety-critical threshold is exceeded (see Safety Manual, Section 3), the equipment must be shut down before further diagnosis, and a technician must confirm isolation before any physical inspection.